

Article in Press

Lml-yolo: Composite enhanced YOLO object detection model for underwater complex environments

Received: 08 Jul 2025

Accepted: 16 Jan 2026

Published online: 20 February 2026

Degang Yang, Chaohao Shen, Tingting Song, Xin Zhang & Zikai Chen

Cite this article as: Yang, D., Shen, C., Song, T. *et al.* Lml-yolo: Composite enhanced YOLO object detection model for underwater complex environments. *J. King Saud Univ. Comput. Inf. Sci.* (2026). <https://doi.org/10.1007/s44443-026-00501-3>

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

LML-YOLO: Composite Enhanced YOLO Target Detection Model for Underwater Complex Environments

Degang Yang¹, Chaohao Shen¹, Tingting Song^{1*}, Xin Zhang², and Zikai Chen¹

¹ College of Computer and Information Science, Chongqing Normal University, Chongqing, 401331, China.

² School of Big Data and Software Engineering, Chongqing University, Chongqing, 400044, China.

Abstract

Underwater target detection encounters technical challenges such as complex background interference, small target scales, random distribution directions, and diverse biomorphology. To address these critical issues, this study proposes LML-YOLO, an enhanced lightweight model based on YOLOv11, which integrates the Lightweight Receptive-Field Attention Convolutional (LRFACnv) module and Multidimensional Directional Collaborative Attention (MDCA). LRFACnv dynamically optimizes convolutional kernel parameters through a receptive-field attention mechanism, effectively enhancing target feature extraction while suppressing background interference. MDCA extends multidimensional spatial collaborative attention by incorporating diagonal and anti-angle directional attention, improving multi-angle feature perception. Additionally, the Neck module uses linear deformable attention convolution (LDAConv), combining deformable convolution with a perceptual field attention mechanism to better adapt to diverse biological morphologies. Experimental results demonstrate that the proposed LML-YOLO achieves mAP scores of 65.2%, 29.8%, and 55% on DUO, UDD, and UODD datasets, respectively, which outperform the original model by 2.1%, 1.5%, and 1.2%, while reducing parameters and GFLOPs by 6.9% and 4.7%. These advancements establish a novel method in efficient underwater target detection, particularly beneficial for marine biological resource monitoring and biodiversity conservation applications.

Corresponding author:

Name: Tingting Song

E-mail: ttsong@cqnu.edu.cn

LML-YOLO: Composite Enhanced YOLO Object Detection Model for Underwater Complex Environments

Abstract

Underwater object detection encounters technical challenges such as complex background interference, small target scales, random distribution directions, and diverse biomorphology. To address these critical issues, this study proposes LML-YOLO, an enhanced lightweight model based on YOLO11n, which integrates the Lightweight Receptive-Field Attention Convolutional (LRFACnv) module and Multidimensional Directional Collaborative Attention (MDCA). LRFACnv dynamically optimizes convolutional kernel parameters through a receptive-field attention mechanism, effectively enhancing target feature extraction while suppressing background interference. MDCA extends multidimensional spatial collaborative attention by incorporating diagonal and anti-angle directional attention, improving multi-angle feature perception. Additionally, the Neck module uses linear deformable attention convolution (LDAConv), combining deformable convolution with a perceptual field attention mechanism to better adapt to diverse biological morphologies. Experimental results demonstrate that the proposed LML-YOLO achieves mAP scores of 65.2%, 29.8%, and 55% on DUO, UDD, and UODD datasets, respectively, which outperform the original model by 2.1%, 1.4%, and 1.2%, while reducing parameters and GFLOPs by 6.9% and 4.7%. These advancements establish a novel method in efficient underwater object detection, particularly beneficial for marine biological resource monitoring and biodiversity conservation applications.

Keywords: Object detection ; Underwater images ; LRFACnv ; MDCA ; LDAConv

1 Introduction

Underwater object detection plays a vital role in marine resource exploration, environmental monitoring, and ecological research. However, it continues to face significant challenges due to the inherent limitations of underwater optical imagery, such as low contrast, blurring, and color distortion [1, 2]. Additionally, targets often vary considerably in size and exhibit complex spatial distributions [3], further complicating detection tasks. Conventional approaches are often constrained by high hardware costs, algorithmic complexity, and substantial computational requirements [4], making it difficult to achieve lightweight yet accurate detection of marine organisms in complex underwater environments.

To improve detection performance, several strategies have been proposed. These include data augmentation techniques [5], multi-scale retinal

color restoration methods [6], and network architecture optimizations [7]. Such methods primarily aim to enhance dataset quality and network capability. Nevertheless, they often fall short in effectively addressing core issues such as background interference and inadequate feature representation for underwater targets. In particular, when dealing with organisms in varied poses or with overlapping structures, these techniques tend to lose fine image details, compromising detection accuracy. Moreover, they frequently exhibit limited robustness and generalization.

In recent years, deep learning-based object detection algorithms have introduced promising alternatives to tackle these problems. These methods have been widely adopted in underwater biological detection and can be broadly categorized into two-stage and single-stage paradigms. Two-stage detectors, such as R-CNN [8] and its variants [9], first generate region proposals and then perform classification and bounding-box

regression. While these models generally achieve high accuracy, they tend to be computationally expensive and lack real-time efficiency. In contrast, single-stage detectors like SSD [10] and the YOLO series [11] eliminate the need for region proposal networks by directly predicting object locations and categories. These offer faster inference speeds and reduced computational overhead, making them more suitable for real-time applications.

Building on the advantages of single-stage detectors, the YOLO series has emerged as a particularly influential framework in underwater environments, prompting numerous improvement studies that have achieved significant progress. Li et al [12] achieved 88.2% accuracy in the scallop detection task by replacing the backbone network of YOLOv5 with a swarm convolution kernel inverted residual block. Shen et al. [13] introduced the mDFLAM module into YOLOv7, employing a multilevel attention mechanism to effectively minimize underwater background interference while enhancing object detection accuracy and maintaining real-time performance. Guo et al [14] enhanced YOLOv8s by adopting a lightweight FasterNet backbone and optimizing the bi-directional feature pyramid network, significantly increasing detection speed. Cen et al. [15] developed YOLOv9-YX, incorporating an enhanced gamma-adaptive contrast-constrained histogram equalization algorithm to improve underwater image quality, along with CDown and FSPPF modules to optimize downsampling and multiscale feature fusion. Tu et al. [16] proposed YOLOv10-UDFishNet, improving YOLOv10 through three key enhancements: auxiliary branching in feature extraction, SPD-Conv replacement for traditional convolution to retain details, and dynamic channel-weight adjustment using high-level semantics, achieving 90.1% recall in diseased triggerfish juvenile detection.

Existing methods have achieved progress in underwater object detection, but challenges remain. First, the complex underwater environment often conceals targets among rocks or corals, while vast backgrounds with color similarities to targets cause significant recognition interference. Second, many detection tasks must operate on resource-limited unmanned platforms like underwater robots, making complex models difficult to

deploy. To address these issues, we propose LML-YOLO, a lightweight underwater biological detection network based on YOLO11n [17], designed to improve detection accuracy while maintaining efficiency for resource-constrained scenarios. The main contributions are as follows:

- We propose LRFAConv, a lightweight receptive-field attention convolution, which overcomes parameter-sharing limitations via receptive-field attention. Further optimized with depth-separable convolution, it enhances biological object detection in complex backgrounds while maintaining model efficiency.
- We propose Multidimensional Directional Collaborative Attention (MDCA), an efficient five-branch module that captures complex multidimensional data dependencies through collaborative attention modeling, achieving significant performance gains without compromising computational efficiency.
- We propose LDAConv, a linear deformable attention convolution that adaptively captures underwater organisms' morphological features via dynamic modulation and X-shaped sampling. By integrating dual-path receptive-field attention, it enables dynamic feature fusion and information compensation, significantly enhancing feature representation.
- Based on the aforementioned improvements, a new underwater object detection model named LML-YOLO was designed, where the two 'L's represent 'LRFAConv' and 'LDAConv', and the 'M' stands for 'MDCA'. Experiments on the DUO, UDD, and UODD datasets demonstrate its outstanding performance.

The structure of the subsequent sections of this paper is summarized as follows: Section 2 outlines convolution and attention mechanisms in object detection. Section 3 details our improvements. Section 4 presents experimental validation and analysis. Section 5 concludes the study.

2 Related work

2.1 Convolution

Downsampling streamlines the image information through convolution operation to reduce the computational cost while retaining the key features, thus improving the model efficiency and

generalization ability and suppressing overfitting. In terms of implementation, Strided Convolution [18] uses a convolution kernel with a step size larger than 1 to reduce the feature map resolution, which can better preserve spatial information compared to pooling operations; Deformable Convolution [19] dynamically adjusts the sampling position to fit the target geometry through learnable offsets; Depthwise Separable Convolution [20] reduces the computational effort and the number of parameters by decomposing the standard convolution into channel-by-channel spatial convolution and 1×1 pointwise convolution; Partial Convolution (PConv) [21] reduces computation and memory access by performing convolution operations on only some of the input channels; RFACnv [22] introduces a receptive-field attention mechanism that breaks parameter-sharing restrictions but increases the computational burden; LDConv [23] can realize convolution kernels with arbitrary number of parameters and sampling shapes, but suffers from the defects of fixed receptive-field and limited sampling amplitude. Based on this, we propose a dual-pathway structure to optimize RFACnv for reduced parameters and computational complexity, while integrating modulation mechanisms and receptive-field attention in LDConv to extend receptive fields, building a more expressive feature extraction framework.

2.2 Attention mechanism

The attention mechanism effectively improves the model's ability to focus on important features by dynamically enhancing key features and suppressing irrelevant information, thus optimizing the model's representation performance. In terms of specific implementation, Squeeze-and-Excitation Networks (SE) [24] adopts the channel attention mechanism to learn inter-channel dependencies, enhance key channel features, and suppress unimportant channel features; Convolutional Block Attention Module (CBAM) [25] cascades the channel and spatial attention modules to achieve dual weighting of channel and space on feature maps; Coordinate Attention (CA) [26] models channel relationships and spatial information embedded simultaneously through coordinate Efficient Channel Attention (ECA) [27] employs 1D convolution to achieve efficient local cross-channel interaction and avoid information loss

due to dimensionality reduction; Mixed Local Channel Attention (MLCA) [28] fuses local and global features through local pooling and inverse pooling operations; and Multidimensional Collaborative Attention (MCA) [29] enhances feature representation by collaboratively modeling channel, height, and width 3D attention. However, traditional horizontal or vertical scanning struggles to capture diagonally distributed features. Thus, we introduce a diagonal and anti-diagonal scanning mechanism based on Multidimensional Collaborative Attention, significantly improving dense object detection accuracy in murky waters through dynamic fusion of multidimensional multidirectional feature relationships.

3 Methodology

3.1 LML-YOLO model

In this study, we construct an underwater object detection model based on YOLO11n, a real-time detection framework developed by Ultralytics. This model significantly improves detection accuracy while maintaining high inference speed through an optimized network architecture and training strategy. To address the challenges of underwater image feature ambiguity and target scale variability, we propose the LML-YOLO model, as shown in Fig. 1. This model achieves performance improvement through three innovative modules: first, the improved LRFAConv module enhances multi-scale feature extraction while reducing parameters via a dual-pathway structure and a receptive-field attention mechanism; second, the integrated MDCA attention mechanism strengthens multi-scale feature extraction through collaborative modeling across multidimensional channel spaces and multi-directional features (including horizontal, vertical, and diagonal elements), thereby effectively enhancing key features and suppressing interference from the underwater environment; finally, the added LDA-Conv module dynamically adapts to the complex geometric shapes of underwater targets through deformable convolution kernels and a modulation mechanism, significantly improving feature representation capability. These improvements enable the model to better cope with complex underwater scenes. The design principles and implementation

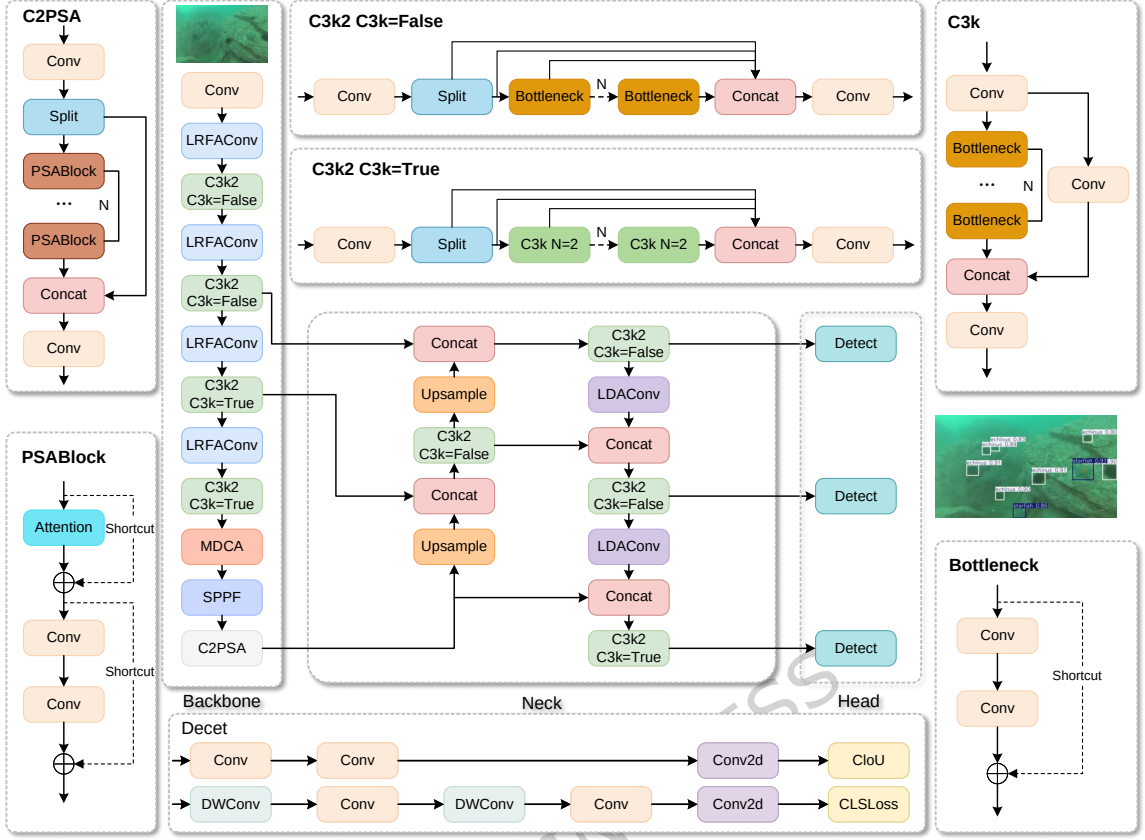


Fig. 1: Schematic diagram of LML-YOLO structure.

methods of each module are described in detail below.

3.2 LRFACConv

Complex underwater images commonly have blurring and uneven lighting. Traditional CNNs, constrained by shared convolution kernel parameters, struggle to precisely handle varying receptive field sizes, limiting target identification and localization. To address this, Zhang et al. [22] combined spatial attention with standard convolution to propose a method called RFACConv. While this method resolves the parameter sharing issue by interacting with receptive field feature information, interacting with each receptive field feature introduces additional computational overhead. This overhead conflicts with the demand for efficient object detection in underwater detectors, as underwater detectors must operate within limited computational resources and storage capacity

while ensuring the accuracy and efficiency of object detection. Therefore, this paper optimizes RFACConv and proposes a lightweight receptive field attention convolution (LRFACConv), whose architecture is shown in Fig. 2.

Specifically, in our implementation, the kernel size k is set to 3 to balance model lightweighting and computational efficiency. The LRFACConv module achieves efficient feature extraction through dynamic weight assignment and multi-branch feature reorganization mechanism. In the weight generation stage, the AvgPool average pooling layer is first used to fuse the $k \times k$ neighborhood information, followed by expanding the channel dimensions to k^2 times by group convolution and assigning dynamic weights to each position in the neighborhood by softmax function. The feature generation module, on the other hand, extends the input features via group convolution (with batch normalization and ReLU activation) into a tensor of dimensions $C \times k^2 \times H \times W$, which

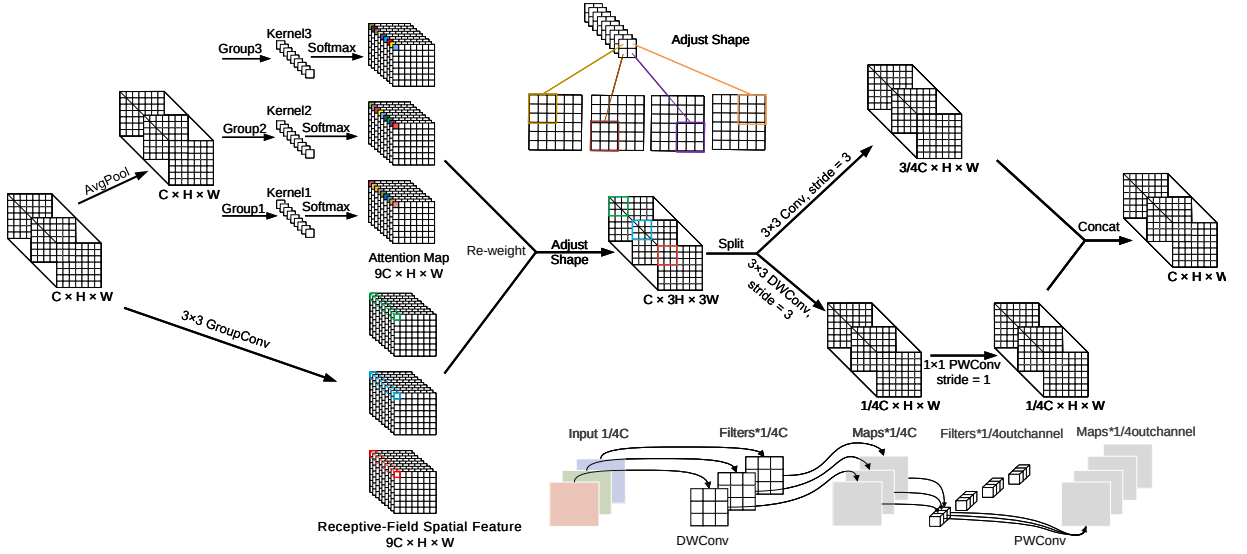


Fig. 2: The structure of LRFAConv.

encodes the information from k^2 spatial locations. The weighted features are restored to the original resolution by tensor reorganization operation and sliced into two groups of feature streams along the channel dimension: the first group reinforces the spatial feature interactions by standard convolution with a step size of 3 for the $\frac{3}{4}C$ channel, and the second group constructs a lightweight branch by using a 3×3 depthwise convolution with a step size of 3 and a 1×1 pointwise convolution for the $\frac{1}{4}C$ channel. Ultimately, through the dual-branch feature fusion mechanism, the number of parameters is reduced while breaking through the weight sharing limitation of traditional convolution, in which the standard convolution branch captures cross-channel spatial correlation and the lightweight branch retains local detailed features, and the two complement each other to form a more discriminative feature expression. The computation of LRFAConv can be described as Eq.(1) and Eq.(2).

Here, $g^{i \times i}$ denotes a grouped convolution of dimension $i \times i$. The parameter k denotes the dimension of the convolution kernel. *Norm* is

an abbreviation for normalization, X denotes the input feature map, and F denotes the re-scaled feature map and the receptive-field attention weighting. Result denotes that the feature map F is split into two parts, $\frac{1}{4}$ -channel and $\frac{3}{4}$ -channel, which are processed by depth-separable convolution and ordinary convolution with a step size of 3, respectively, and finally the final result is output by channel merging.

3.3 MDCA

Organisms in underwater images are arbitrarily distributed, making traditional horizontal or vertical scanning often inadequate for capturing obliquely oriented features. To address this, we propose a multi-dimensional, multi-directional attention module named MDCA. It dynamically captures feature interactions by synchronizing channel and spatial distribution analysis, and fuses the multidimensional information to enhance detection accuracy for dense targets in turbid waters.

$$F = \text{Softmax} \left(g^{i \times i} (\text{AvgPool}(X)) \right) \times \text{ReLU} \left(\text{Norm} \left(g^{k \times k} (X) \right) \right) \quad (1)$$

$$\text{Result} = \text{Cat} \left(\text{PWConv}_{1 \times 1} \left(\text{DWConv}_{3 \times 3} \left(F_{\frac{1}{4}C} \right) \right), \text{Conv}_{3 \times 3} \left(F_{\frac{3}{4}C} \right) \right) \quad (2)$$

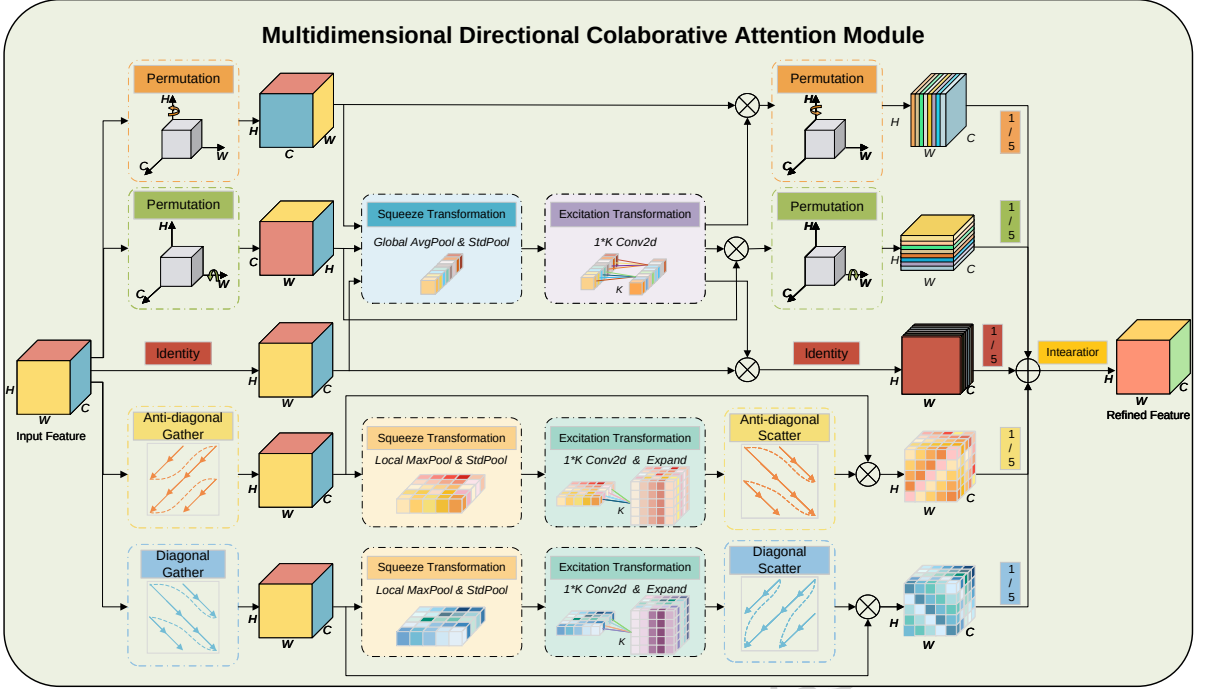


Fig. 3: MDCA overall architecture diagram.

The MDCA we propose incorporates a multi-dimensional collaborative attention mechanism, with its overall architecture shown in Fig. 3. This module integrates feature mapping relationships through five parallel branches: the first and second branches capture width and height features in the spatial dimension, respectively; the third branch focuses on channel-to-channel interactions; the fourth and fifth branches enhance diagonal and anti-diagonal scans in the spatial dimension. This design not only reinforces the original multidimensionality but also introduces multi-directional attention, effectively capturing the complex dependencies of input data across multiple dimensions and directions. This enhances the model's focus on key features and provides a more comprehensive coverage of the spatial distribution of underwater organisms, thereby significantly improving the model's recognition performance.

Specifically, our proposed MDCA takes the feature map $F \in \mathbb{R}^{C \times H \times W}$ as input and first transforms it into three different dimensional representations through rotation and identity mapping. These representations correspond to three

distinct branches: $\hat{F}_W \in \mathbb{R}^{W \times H \times C}$ denotes the feature map from the first branch, obtained by rotating the input counterclockwise about the W -axis; $\hat{F}_H \in \mathbb{R}^{H \times C \times W}$ corresponds to the output of the second branch rotated counterclockwise about the H -axis; and $\hat{F}_C \in \mathbb{R}^{C \times H \times W}$ represents the original feature map retained through identity mapping in the third (channel) branch.

$$\hat{F}_W = PM_H(F) \quad (3)$$

$$\hat{F}_H = PM_W(F) \quad (4)$$

$$\hat{F}_C = IM(F) \quad (5)$$

Here, PM_H denotes the 90° counterclockwise rotation along the H -axis, PM_W denotes the 90° counterclockwise rotation about the W -axis, and IM indicates the identity mapping operation.

In each branch, we apply both global average pooling and global standard deviation pooling (as computed in Eq.(6) and Eq.(7)) to capture first- and second-order statistics over their respective

spatial-channel dimensions.

$$\hat{f}_m^{\text{avg}} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \hat{f}_m(i, j) \quad (6)$$

$$\hat{f}_m^{\text{std}} = \sqrt{\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (\hat{f}_m(i, j) - \hat{f}_m^{\text{avg}})^2} \quad (7)$$

Here, $\hat{f}_m \in \mathbb{R}^{1 \times H \times W}$ denotes the m -th channel feature map of the input feature map $\hat{F}_k$. $\hat{f}_m^{\text{avg}}$ and $\hat{f}_m^{\text{std}}$ correspond to the two feature descriptors of the channel.

The G_{sq} compression transform of the mean and standard deviation statistics generates $\bar{F}_W \in \mathbb{R}^{W \times 1 \times 1}$ and $\bar{F}_H \in \mathbb{R}^{H \times 1 \times 1}$, respectively, and then outputs the weights in the width and height directions after the G_{ex} transform of the convolutional kernel size $(1, K)$. These weights are combined with $A_W \in \mathbb{R}^{W \times 1 \times 1}$ and $A_H \in \mathbb{R}^{H \times 1 \times 1}$ generated by the Sigmoid function. Element-by-element multiplication enhances the feature maps to obtain $F'_W \in \mathbb{R}^{W \times H \times C}$ and $F'_H \in \mathbb{R}^{H \times C \times W}$. To make the formulas concise, we use the symbols $t \in \{H, W, C\}$ to refer to H , W , and C to avoid repetitive expressions.

$$G_{\text{sq}}(\cdot) = \frac{1}{2} \otimes (\hat{F}_t^{\text{avg}} \oplus \hat{F}_t^{\text{std}}) \oplus \alpha \otimes \hat{F}_t^{\text{avg}} \oplus \beta \otimes \hat{F}_t^{\text{std}} \quad (8)$$

$$\bar{F}_t = G_{\text{sq}}(\hat{F}_t) \quad (9)$$

$$\tilde{F}_t = G_{\text{ex}}(\bar{F}_t) \quad (10)$$

$$A_t = \sigma(\tilde{F}_t) \in [0, 1] \quad (11)$$

$$F'_t = A_t \otimes \tilde{F}_t \quad (12)$$

Here, $\sigma(\cdot)$ denotes the Sigmoid activation function. α and β are two trainable parameters constrained to the range $(0, 1)$, which are optimized through stochastic gradient descent (SGD). Accordingly, our method adaptively assigns distinct weights to average-pooled, max-pooled, and standard-deviation-pooled features at various stages of image feature extraction.

The enhanced features F'_W and F'_H are then rotated back to the original layout via inverse permutation, as expressed by the following formula:

$$F''_W = PM_H^{-1}(F'_W) \quad (13)$$

$$F''_H = PM_W^{-1}(F'_H) \quad (14)$$

$$F'_C = IM(F'_C) \quad (15)$$

Here, PM_H^{-1} denotes the 90° clockwise rotation about the H -axis, PM_W^{-1} denotes the 90° clockwise rotation about the W -axis.

In the fourth and fifth branches of the MDCA module, feature maps are modeled along the main diagonal and antidiagonal directions as shown in Fig. 4. For a single-channel feature map $I \in \mathbb{Z}^{H \times W}$, the Gather operation is defined via the index mappings:

$$I_D[i, j] = (j - i) \% W \quad (16)$$

$$I_{AD}[i, j] = (i + j) \% W \quad (17)$$

$$\hat{X}_D = \text{Gather}(F, I_D) \quad (18)$$

$$\hat{X}_{AD} = \text{Gather}(F, I_{AD}) \quad (19)$$

These mappings enable shifting the feature values along the specified directions. Subsequently, for each position $j \in [1, W]$ of the m -th channel, two statistical measures—maximum value $\hat{g}_m^{\text{max}}(j)$ and standard deviation $\hat{g}_m^{\text{std}}$ are computed along the height dimension via the L_{sq} compression transform to capture local feature information:

$$\hat{g}_m^{\text{max}}(j) = \max_{i \in [1, H]} g_m(i, j), \quad \forall j \in [1, W] \quad (20)$$

$$\hat{g}_m^{\text{std}}(j) = \sqrt{\frac{1}{H} \sum_{i=1}^H (g_m(i, j) - \hat{g}_m^{\text{avg}}(j))^2} \quad (21)$$

These statistics are fused by weighting and then transformed by L_{ex} excitation with convolutional kernel size $(1, K)$ to obtain $\tilde{X}_T$, which is augmented by element-by-element multiplication of the feature maps with A_T generated by the Sigmoid function to obtain X'_D and X'_{AD} . In order to make the formula concise, we use the symbol $T \in \{D, AD\}$ to refer to the diagonal D and the antidiagonal AD to avoid repetitive expressions, and the expression of this part of the formula is shown below:

$$L_{\text{sq}}(\cdot) = \frac{1}{2} \otimes (\hat{X}_T^{\text{Max}} \oplus \hat{X}_T^{\text{Std}}) \oplus \alpha \otimes \hat{X}_T^{\text{Max}} \oplus \beta \otimes \hat{X}_T^{\text{Std}} \quad (22)$$

$$\bar{X}_T = L_{\text{sq}}(\hat{X}_T) \quad (23)$$

$$\tilde{X}_T = L_{\text{ex}}(\bar{X}_T) \quad (24)$$

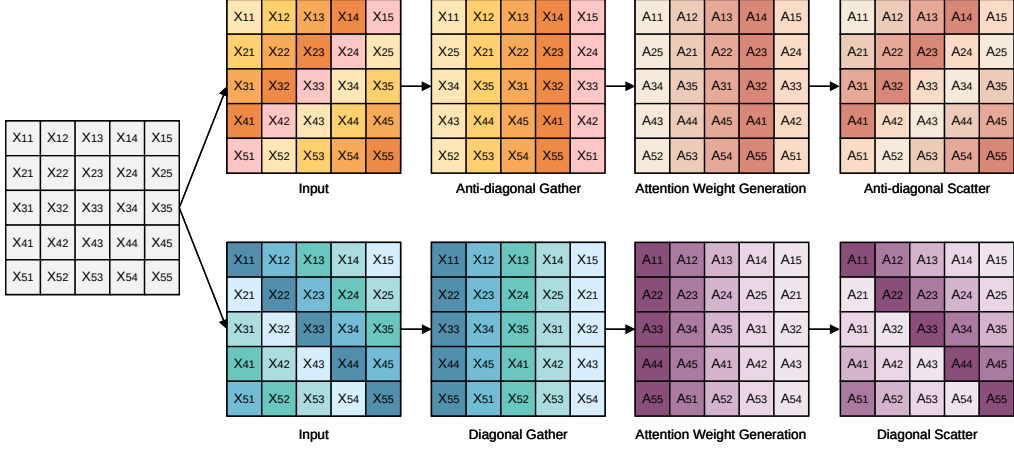


Fig. 4: Diagonal and antidiagonal acquisition and scattering schematics.

$$A_T = \sigma(\tilde{X}_T) \in [0, 1] \quad (25)$$

$$X'_T = A_T \otimes \tilde{X}_T \quad (26)$$

The enhanced weights are accurately reduced to the original diagonal and antidiagonal positions by the inverse index I_D^{-1} and I_{AD}^{-1} -based scatter operation, and finally multiplied element by element with the input features to realize the oblique enhancement of local features.

$$X''_D = \text{Scatter}(X'_D, I_D^{-1}) \quad (27)$$

$$X''_{AD} = \text{Scatter}(X'_{AD}, I_{AD}^{-1}) \quad (28)$$

Here, Gather and Scatter can be implemented through the PyTorch toolbox [30].

In the final fusion stage of MDCA, the output features of all branches are integrated by an equal weight fusion strategy. This strategy significantly enhances the model's ability to capture orientation-sensitive features such as image edges and textures by balanced aggregation of height, width, channel, diagonal and anti-diagonal directions. In complex scenes such as fish occlusion or turbid waters, this multidimensional and multidirectional complementary mechanism can effectively alleviate the feature confusion problem caused by target overlapping.

$$F'' = \frac{1}{5} \otimes (F''_W \oplus F''_H \oplus F''_C \oplus F''_D \oplus X''_{AD}) \quad (29)$$

3.4 LDAConv

Underwater organisms vary in shape and size, while standard convolution uses fixed, typically square kernels. This may struggle to efficiently capture features of elongated fish or round sea urchins. To address this, we design a Linear Deformable Attention Convolution (LDAConv) to replace standard convolution; its overall structure is shown in Fig. 5.

In this module, we integrate the dynamic modulation mechanism of the Deformable Convolution v2 [31], which not only learns the offset of the sampling position, but also dynamically adjusts the feature amplitude of each sampling point by the modulation scalar, so as to control the degree of contribution of the spatial support region to the features more precisely.

In the modulation mechanism, the weight for each sampling point can be described as follows: Let $y(p)$ be the output feature map value at position p . Here, N represents the number of sampling points in the convolution kernel, and w_n denotes the weight at the n -th position of the kernel. Given the input feature map x , where p is a position on the feature map and p_n is the predefined offset at the n -th position of the kernel, the learnable offset and the modulation scalar at the n -th position are denoted by Δp_n and Δm_n , respectively.

$$y(p) = \sum_{n=1}^N w_n \cdot x(p + p_n + \Delta p_n) \cdot \Delta m_n \quad (30)$$

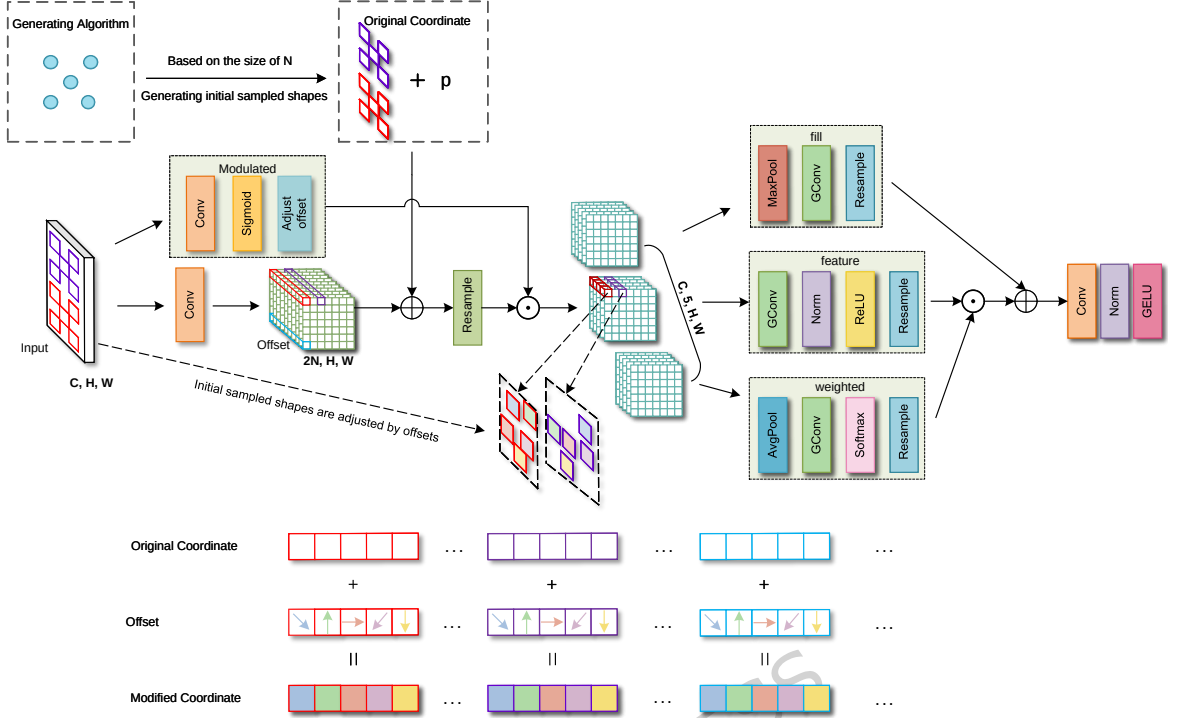


Fig. 5: Overall structure of LDAConv.

According to Eq.(30), the convolutional kernel size N in LDAConv can be configured with different dimensions such as 1×3 or 1×5 , while various initial sampling shapes p_n (e.g., X-shaped or H-shaped) can be specified. These shapes support configurable horizontal and vertical spacings between sampling points on the feature map—for instance, uniform spacings of 2 pixels or 4 pixels in both directions. This flexibility in convolutional kernel quantity and initial sampling configuration allows the model to adapt to diverse datasets. As shown in Table 5 and Fig. 8, given the high variability in organism morphology and size in underwater environments, we compared multiple initial sampling shapes and spacing configurations, and ultimately selected an X-shaped structure with a uniform spacing of 4 pixels in both directions as the initial sampling setup.

In addition, we incorporate receptive-field attention in LDAConv for dynamic feature fusion and information compensation. Specifically, the mechanism contains a dual-pathway design: the

dynamic weighting pathway uses average pooling to capture 3×3 neighborhood information, extends the input channel to n times the number of sampling points by group convolution, and generates normalized weights along the channel dimensions using the Softmax function to achieve spatially adaptive feature importance assignment; and the information compensation pathway extracts locally salient features through max pooling and uses group convolution to expand the channel dimension, generating supplemental features to alleviate the information loss in the sampling process. After dual-pathway output, the model convolves, normalizes, and activates bilinearly interpolated offset features to enhance representation. These features are then element-wise multiplied with dynamic weights to highlight key regions. Finally, weighted features fuse with complementary features for synergistic enhancement of feature expression. The receptive field space transformations are illustrated in Fig. 6.

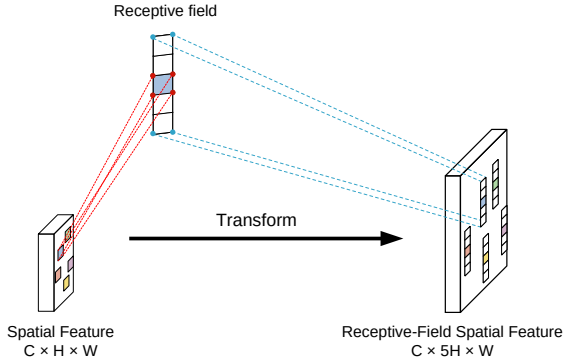


Fig. 6: Feeling field spatial features obtained by transforming spatial features.

4 Experiments and analysis

In this study, we validate the performance of the LML-YOLO model through two sets of systematic experiments: in the experimental design section, we firstly use the DUO [32] dataset for ablation experiments, focusing on verifying the architectural efficiency of the three core modules, namely, LRFACnv, MDCA, and LDAConv; and subsequently, we use the two publicly available datasets, namely, UDD [33] and UODD [34], to construct a common with the DUO comparison experimental group to comprehensively evaluate the comprehensive performance of the model in underwater bio-detection tasks.

4.1 Datasets

DUO: The Detecting Underwater Objects (DUO) dataset, proposed by the International School of Information Science and Engineering, DUTRU, Dalian University of Technology, includes 7,782 deduplicated and diverse underwater images—6,671 for training and 1,111 for testing. It covers four target types: holothurian, echinus, scallop, and starfish. Most targets occupy less than 1.5% of the image area, averaging 9.57 instances per image.

UDD: The Underwater Open-sea Farm Object Detection Dataset (UDD), introduced by Dalian University of Technology’s Multimedia Laboratory, consists of 2,227 images (1,827 for training, 400 for validation). It features sea cucumbers, sea urchins, and scallops from real

sea farms, with densely populated small targets accounting for only 0.47% of the image area.

UODD: The Underwater Object Detection Dataset (UODD), jointly developed by Dalian University of Technology and ByteDance, contains 3,194 real underwater scene images—2,560 for training, 128 for validation, and 506 for testing. Relabeled from the RUIE dataset, it includes holothurian, sea echinus, and scallop, with numerous small targets averaging about 6 instances per image.

4.2 Experimental setting

To ensure model comparability, the experimental environment configuration is detailed in Table 1. Each dataset underwent 200 training epochs (batch size of 16), with hyperparameters such as learning rate and regularization strength maintained at default settings throughout training to ensure consistency.

Table 1: Experimental configuration.

Configure	Version
Operating system	Windows 10
CPU	Intel i5-12490F
GPU	GeForce RTX 4060Ti
Python	3.8.18
Optimizer	SGD
Framework	Pytorch 2.0.1
Accelerated environment	CUDA 11.7

4.3 Evaluation metrics

During experiments, we evaluated algorithm performance using three common metrics: Precision, Recall, and Mean Average Precision (mAP). Precision quantifies the proportion of correctly identified positives among detected samples; Recall measures the proportion of actual positives successfully detected. mAP provides a comprehensive evaluation by averaging the Average Precision (AP) values across all classes, where each AP corresponds to the area under the precision-recall curve for a specific category. In this study, we report two standard variants: mAP50, which assesses detection quality at a single IoU threshold of 0.5, and the overall mAP (averaged over 10 IoU thresholds from 0.5 to 0.95 in 0.05 increments),

Table 2: Results of ablation experiments on the DUO dataset.

Method				Precision(%)	Recall(%)	mAP50(%)	mAP(%)	GFLOPs	Params(M)	FPS
YOLO11n	LRFAConv	MDCA	LDAConv							
✓	-	-	-	83.7	73.7	82.9	63.1	6.3	2.58	454.54
✓	✓	-	-	84.9	74.6	83.7	64.2	6	2.41	384.61
✓	-	✓	-	85.2	73.9	83.7	64.2	6.3	2.58	588.23
✓	-	-	✓	83.6	76.7	84.1	64.5	6.3	2.57	476.19
✓	✓	✓	-	86.6	75.7	84.1	64.6	6	2.51	344.82
✓	✓	-	✓	85.8	74.8	84.3	64.8	6	2.4	285.71
✓	✓	✓	✓	84.2	77.2	84.8	65.2	6	2.4	285.71

reflecting performance across varying localization. The specific formulas are as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (31)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (32)$$

$$\text{AP} = \int_0^1 \text{Precision}(\text{Recall}) d_{\text{Recall}} \quad (33)$$

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (34)$$

In this paper, we also evaluate the dimensionality and scale of the model through Parameters (Params), Floating Point Operations (GFLOPs), and Frames Per Second (FPS). Here, Parameters refer to the sum of all trainable weights and biases in the model, directly reflecting the model’s scale and complexity; GFLOPs is used to measure computational performance, estimating the computational overhead during the inference and training phases; while FPS indicates the inference speed of the model in deployment scenarios.

4.4 Ablation experiment

The LML-YOLO detection model enhances underwater bio-detection performance by integrating three core modules: LRFAConv, MDCA, and LDAConv. Ablation experiments on the DUO dataset confirm its balanced optimization of accuracy and efficiency, as shown in Table 2.

The experimental results show that the baseline model improves mAP50 and mAP to 83.7% and 64.2%, respectively, with 4.8% less GFLOPs and 6.6% fewer parameters, while the precision

and recall are improved by 1.2% and 0.9%, respectively, with the application of the LRFAConv module to the main trunk. Despite the decrease in FPS, this demonstrates that LRFAConv effectively enhances the model’s feature extraction capabilities for blurred and occluded underwater targets through its receptive field attention mechanism, at an acceptable computational cost. Subsequently, adding a layer of MDCA attention to capture the information of the input data in multiple dimensions and directions resulted in a 2.9% and 2% improvement in detection precision and recall, and the mAP50 and mAP were improved to 84.1% and 64.6%, respectively. Finally, LDAConv is referenced to the neck, which effectively responds to underwater organisms of different sizes and shapes by combining deformable sampling points with sensory field attention, which in turn improves the recall by 3.5% and results in an improvement of 1.9% and 2.1% for mAP50 and mAP, respectively, with respect to the baseline model, and effectively improves the overall generalization ability of the model, with the integrated modules demonstrating a deliberate precision-recall trade-off that prioritizes recall to minimize false negatives for challenging underwater samples while maintaining acceptable precision.

To analyze the impact of the dual-path structure in LRFAConv on model performance under different channel allocation scenarios, we conducted five sets of ablation experiments with varying channel counts, as shown in Table 4. Experiments demonstrate that when the standard convolution path is configured with 3C/4 channels and the depthwise separable convolution path is set to C/4 channels, LRFAConv achieves performance comparable to a single-path architecture based entirely on standard convolutions while

Table 3: Performance comparison of MDCA and various attention mechanisms on the DUO dataset, where AP_S , AP_M , and AP_L represent the detection accuracy for small, medium, and large objects.

Attention	P(%)	R(%)	mAP50(%)	Category				mAP(%)	AP_S (%)	AP_M (%)	AP_L (%)	GFLOPs	Params(M)
				Holothurian	Echinus	Scallop	Starfish						
CA [26]	84.4	74.7	83.4	59.8	73.8	46.8	73.0	63.3	43.5	61.6	57.6	6.3	2.58
CBAM [25]	85.7	73.2	83.2	59.7	74.0	44.9	72.9	62.9	38.2	61.0	57.5	6.3	2.64
CPA [35]	84.1	73.7	82.9	60.0	73.7	44.7	73.3	62.9	42.3	60.9	57.2	6.5	2.71
ECA [27]	83.4	75.2	82.9	60.0	74.0	45.3	73.0	63.1	50.0	61.3	57.1	6.3	2.58
EMA [36]	84.6	75.3	83.4	61.6	73.7	45.9	73.2	63.6	46.0	61.9	58.2	6.4	2.59
LSAK [37]	84.2	73.9	82.8	61.2	73.8	45.2	73.1	63.3	45.6	61.3	57.7	6.4	2.65
MCA [29]	84.4	74.1	82.7	60.6	73.6	45.0	73.3	63.1	47.9	61.2	57.2	6.3	2.58
MLCA [29]	82.4	74.8	83.2	59.4	73.7	45.2	73.0	62.8	38.6	60.8	57.0	6.3	2.58
SA [38]	83.0	75.2	83.3	60.9	73.9	45.5	73.5	63.4	39.8	61.4	57.3	6.3	2.58
SE [24]	84.6	73.9	82.9	59.6	74.0	45.6	73.3	63.1	44.8	61.1	57.1	6.3	2.58
MDCA(ours)	85.2	73.9	83.7	60.9	74.2	48.3	73.6	64.2	39.7	62.0	59.1	6.3	2.58

Table 4: LRFACnv dual-path channel splitting ratio ablation experiment (C: input channel count, (0,C): grouping example, where the standard convolution path has 0 channels and the depth-separable convolution path has C channels)

Split Ratio	mAP50(%)	mAP(%)	Params(M)	GFLOPs
(0, C)	83.0	63.1	2.193	5.4
(C/4, 3C/4)	82.8	62.0	2.198	5.4
(C/2, C/2)	83.7	63.3	2.271	5.6
(3C/4, C/4)	83.7	64.2	2.411	6.0
(C, 0)	83.8	64.5	2.619	6.6

significantly reducing parameter count and computational complexity. This configuration delivers superior efficiency while maintaining competitive performance.

To validate the effectiveness of the MDCA attention mechanism in underwater detection, we conducted comparative experiments with several mainstream attention mechanisms. All experiments were conducted under uniform settings, with each attention module inserted into the baseline network at the same position as the MDCA module in LML-YOLO (as shown in Fig. 1). Experimental results are presented in Table 3. Additionally, to provide more intuitive analysis, LayerCAM visualization was employed in this study. LayerCAM clearly displays the regions of interest for target features at different network levels, revealing how the model leverages contextual information and local details. We randomly selected images from the validation set and visualized comparisons between the original YOLO11n model and the YOLO11n model with MDCA attention. As shown in Fig. 7, compared

to the original architecture, the model incorporating MDCA attention more accurately focuses on the effective regions of the target. This demonstrates that MDCA enhances the model’s ability to extract and utilize key features of the target.

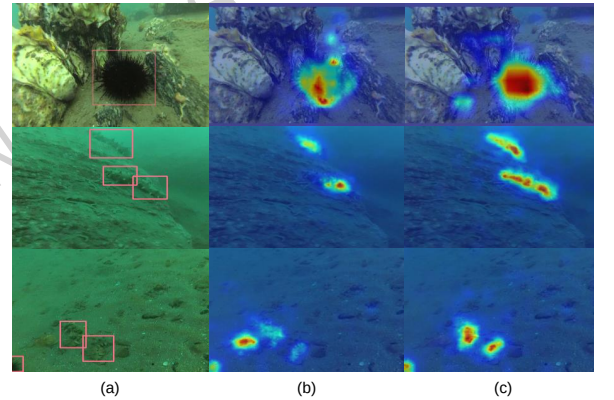
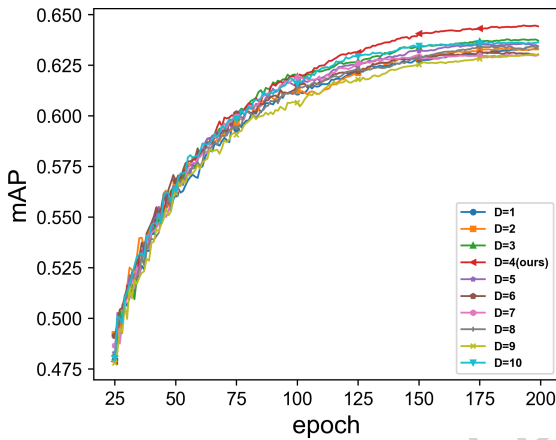


Fig. 7: Heatmap comparison: (a) Input image; (b) Heatmap of the original YOLO11n network; (c) Heatmap of YOLO11n using MDCA. In (b) and (c), a hue closer to red indicates higher attention from the network to that region.

This study compares the performance of multiple LDAConv initial sampling shapes for underwater detection, as shown in Table 5. Experiments show that LDAConv obtains the highest mAP while keeping the number of parameters and GFLOPs constant when the sampling size is 5 and the shape is X. Further studies find that the spacing of the sampling points has a significant effect on the detection results. As demonstrated

Table 5: Comparison of the effects of different initial sampling shapes for LDAConv on the DUO dataset.

Initial Shapes	P(%)	R(%)	mAP50(%)	Category				mAP(%)	AP_S (%)	AP_M (%)	AP_L (%)	GFLOPs	Params(M)
				Holothurian	Echinus	Scallop	Starfish						
	86.1	73.5	82.9	59.9	73.8	45.6	73.3	63.2	43.3	61.2	57.1	6.2	2.45
	85.2	74.6	83.1	60.2	74.3	44.5	73.4	63.1	41.4	61.3	56.7	6.2	2.49
	83.2	74.9	83.2	60.5	74.1	45.7	73.0	63.3	40.6	61.6	57.4	6.3	2.53
	84.8	75.8	83.7	60.3	74.1	47.7	73.0	63.8	42.5	61.9	58.6	6.3	2.57
	85.5	74.6	83.7	60.4	73.8	46.8	72.6	63.4	38.3	62.4	57.4	6.4	2.61
	83.4	75.5	83.3	60.1	73.6	47.4	72.8	63.5	42.9	61.7	58.1	6.5	2.66

**Fig. 8:** Effect of pixel spacing (D) variation between sampling points in initial X-shaped sampling on LDAConv performance.

in Fig. 8, by testing different horizontal and vertical spacing from 1 to 10, the optimal performance was found for spacing $D = 4$. Therefore, the final LDAConv in this paper adopts the configuration of size 5, shape X, and spacing 4.

4.5 Comparison experiment

To evaluate the performance of the proposed algorithm, comparative experiments are conducted on the DUO, UDD, and UODD datasets. To further demonstrate its effectiveness in detecting underwater biological objects, our model is compared against several common detection models. Among them, results for models marked "*" are sourced from the comparative study on underwater detection presented in paper [39].

In the comparison experiments on the DUO, UDD and UODD datasets, LML-YOLO demonstrates an excellent balance of performance and efficiency. As shown in Table 6, the mAP50 of this model on the DUO, UDD and UODD datasets reaches 84.8%, 68.7% and 90.7%, respectively, with a better average accuracy than the other comparative models; the mAP reaches 65.2%, 29.8% and 55%, respectively, and when compared with the benchmark model, YOLO11n, LML-YOLO improves the mAP by 2.1%, 1.4% and 1.2% and also reduces the number of parameters and computational complexity by 6.9% and 4.7%, respectively. The experimental results indicate that LML-YOLO successfully achieves a precise balance between detection accuracy and computational efficiency while maintaining a lightweight architecture.

4.6 Detection results comparison

A comparative analysis of the detection results reveals that severe blurring and background interference in underwater scenes pose significant challenges for object detection. Fig. 9, Fig. 10 and Fig. 11 demonstrate the detection performance of YOLO11n and LML-YOLO on the DUO, UDD, and UODD datasets, respectively. In the examples shown in the first column, LML-YOLO achieves higher detection accuracy for sea urchins, sea cucumbers, and scallops. In the second and third columns, YOLO11n produces notable false positives—such as misclassifying ropes and rocks as sea cucumbers, or identifying blue pipes and rocks as starfish and shells—while LML-YOLO effectively suppresses these errors. In dimly lit

Table 6: Comparison of object detection performance across different models on the DUO, UDD, and UODD datasets.

Models	mAP50 (%)			mAP (%)			GFLOPs	Params (M)
	DUO	UDD	UODD	DUO	UDD	UODD		
Faster R-CNN [40]	75.9	60.3	86.5	54.8	25.7	47.1	63.26	41.14
Cascade R-CNN [41]	75.5	65.1	86.4	55.6	28.0	48.5	77.54	68.94
RT-DETR-L [42]	80.0	25.6	78.9	57.9	11.5	43.5	100.6	28.45
RT-DETR-R50 [42]	83.6	33.1	72.5	60.9	14.9	39.3	125.6	41.94
D-FINE-N [43]	37.4	38.7	62.4	28.7	17.2	34.9	7.12	3.72
Boosting R-CNN* [44]	80.6	64.3	88.0	60.8	28.4	50.5	54.71	45.95
RoIAttn* [45]	79.5	64.5	86.3	58.7	28.3	47.6	331.39	55.23
RoIMix* [5]	80.0	64.9	87.0	61.0	28.0	49.2	91.08	68.94
GCC-Net* [46]	81.6	64.6	87.0	61.1	26.3	46.9	78.93	38.31
RFTM* [47]	79.4	66.1	87.0	60.1	28.8	49.7	91.06	75.58
ERL-Net* [48]	81.4	67.4	88.0	61.2	29.6	49.4	416.63	218.83
YOLOv5 [49]	83.0	63.1	89.0	63.2	27.2	52.7	7.1	2.50
YOLOv6 [50]	82.3	68.5	88.5	62.6	28.8	52.7	11.8	4.23
YOLOv7-tiny [51]	84.8	64.5	90.2	61.8	24.7	50.8	13.2	6.02
YOLOv8n [52]	83.5	63.0	88.6	63.7	26.3	53.3	8.1	3.00
YOLOv9t [53]	82.5	66.8	87.3	63.3	28.6	51.5	6.4	1.73
YOLOv10n [54]	80.9	64.9	88.9	61.8	27.4	54.3	8.2	2.69
YOLO11n [17]	82.9	66.2	89.5	63.1	28.4	53.8	6.3	2.58
YOLO12n [55]	82.4	67.3	87.6	62.2	27.9	51.6	5.8	2.51
YOLOv13n [56]	82.8	61.0	86.4	62.3	25.8	51.1	6.2	2.45
RGM-YOLO [57]	83.5	66.1	89.8	64.0	29.8	54.7	7.1	1.89
LML-YOLO (Ours)	84.8	68.7	90.7	65.2	29.8	55.0	6.0	2.40

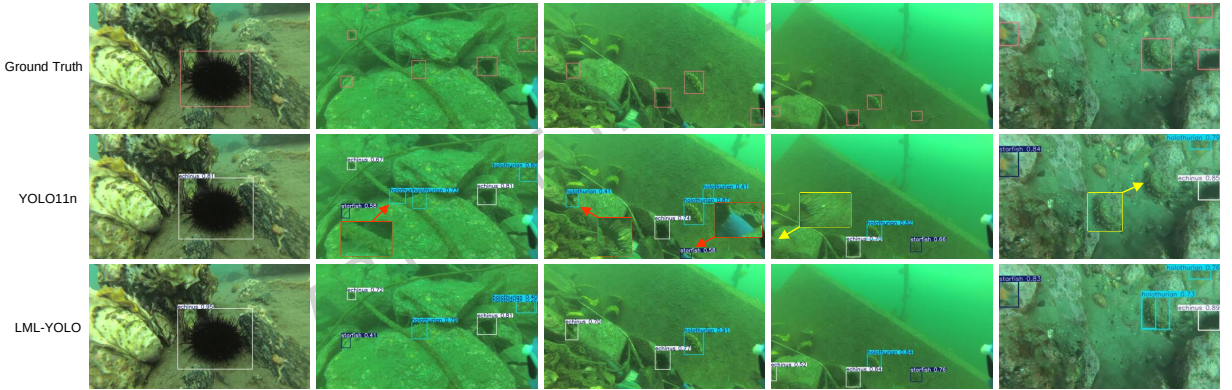


Fig. 9: Comparison of the effects on the DUO dataset. The red and yellow arrows in the figure indicate False Positive (FP) and False Negative (FN), respectively.

scenes with blurred backgrounds (fourth and fifth columns), YOLO11n frequently fails to detect sea cucumbers, whereas LML-YOLO maintains more complete detection capabilities.

Notably, during visualization, we identified extremely rare omissions in the official dataset annotations. For instance, in the fifth column of Fig. 10, two sea cucumbers that should have been annotated were missing from the ground truth labels, yet LML-YOLO correctly detected this

unannotated target. This further demonstrates the model’s strong generalization ability. However, LML-YOLO still exhibits occasional false negatives (e.g., the partially occluded starfish in the third column of Fig. 9). Nevertheless, compared to the baseline model YOLO11n, which not only fails to detect this object but also generated false positives in the same image, the proposed model shows superior overall performance, exhibiting greater robustness and accuracy.

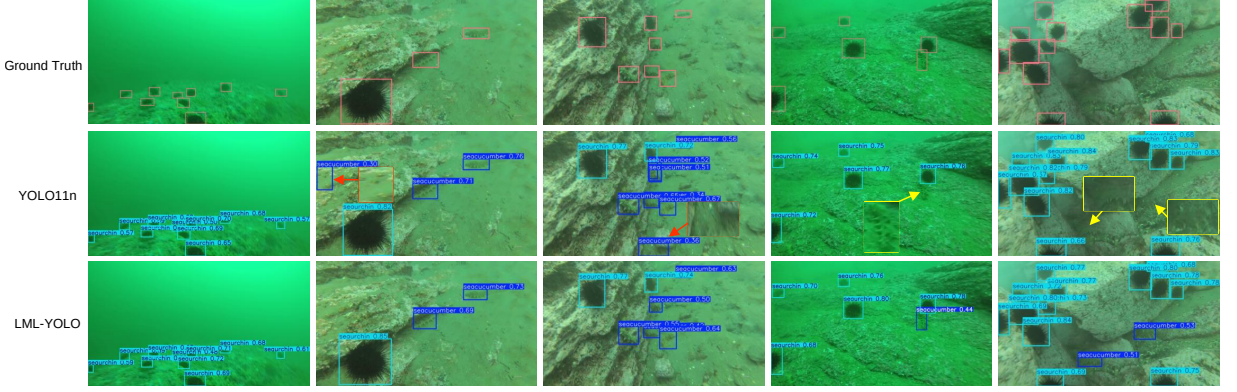


Fig. 10: Comparison of the effects on the UDD dataset. The red and yellow arrows in the figure indicate FP and FN, respectively.

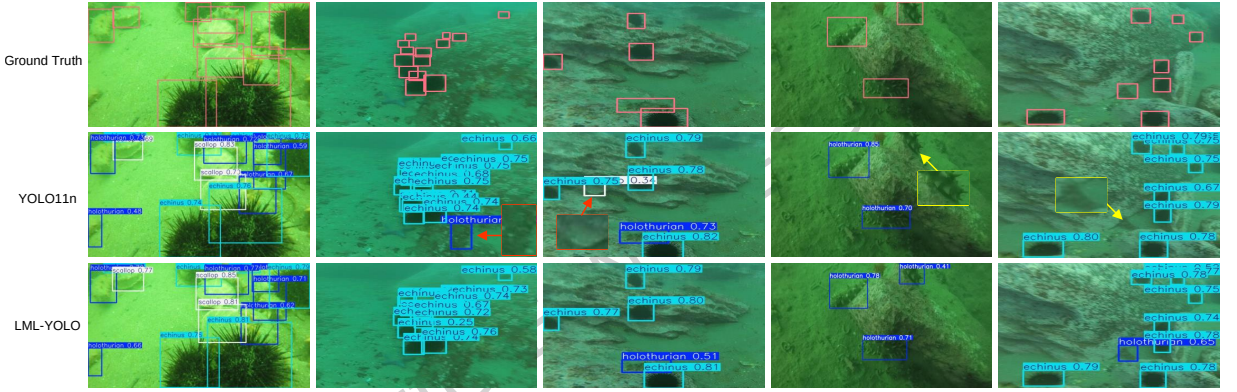


Fig. 11: Comparison of the effects on the UODD dataset. The red and yellow arrows in the figure indicate FP and FN, respectively.

5 Conclusions

This study has developed a lightweight detection model LML-YOLO based on the YOLO11n framework to address challenges including underwater complex background interference, multidirectional target distribution, and morphological diversity. Three core innovations were achieved: firstly, the LRFAConv module enabled efficient feature extraction through a dual-pathway structure with receptive-field attention; secondly, the MDCA mechanism enhanced oblique feature capture capability by merging channel-spatial dimensions with multidirectional scanning; finally, the

LDACConv module improved adaptability to heterogeneous targets through deformable convolution and modulation mechanisms. Experimental validation on DUO, UDD, and UODD datasets has demonstrated significantly improved detection accuracy while maintaining model lightweightness. The achieved balance between accuracy and efficiency makes this model applicable for underwater detection tasks such as marine survey and environmental monitoring. Nevertheless, detection speed remains an aspect for further improvement, and future efforts will focus on optimizing the model's real-time performance.

Declarations

Availability of data and material The DUO, UDD and UODD datasets used in this paper are open and can be downloaded from the Internet. All data included in this study are available upon request by contact with the corresponding author.

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding This research was supported in part by the Science and Technology Research Program of Chongqing Municipal Education Commission (KJQN202200537 and KJZD-M202300502), in part by Chongqing Normal University Ph.D. Start-up Fund (21XLB035).

Authors' contributions XXXXXX XXXX: Conceptualization, Methodology, Resources, Writing-review & editing, Project administration, Supervision, Visualization, Funding acquisition. XXXXXXXX XXXX: Conceptualization, Methodology, Validation, Software, Formal analysis, Investigation, Writing-original draft, Writing-review & editing, Visualization. XXXXXXXX XXXX: Conceptualization, Methodology, Resources, Writing-review & editing, Project administration, Supervision, Visualization, Funding acquisition. XXX XXXX: Data Curation, Visualization, Formal analysis, Validation. XXXXX XXXX: Data Curation, Visualization, Formal analysis, Validation.

References

1. Xu, S., Zhang, M., Song, W., Mei, H., He, Q., Liotta, A.: A systematic review and analysis of deep learning-based underwater object detection. *Neurocomputing* **527**, 204–232 (2023). <https://doi.org/10.1016/j.neucom.2023.01.056>
2. Zhang, W., Zhuang, P., Sun, H.-H., Li, G., Kwong, S., Li, C.: Underwater image enhancement via minimal color loss and locally adaptive contrast enhancement. *IEEE Transactions on Image Processing* **31**, 3997–4010 (2022). <https://doi.org/10.1109/TIP.2022.3177129>
3. Jian, M., Yang, N., Tao, C., Zhi, H., Luo, H.: Underwater object detection and datasets: a survey. *Intelligent Marine Technology and Systems* **2**(1), 9 (2024). <https://doi.org/10.1007/s44295-024-00023-6>
4. Cong, Y., Gu, C., Zhang, T., Gao, Y.: Underwater robot sensing technology: A survey. *Fundamental Research* **1**(3), 337–345 (2021). <https://doi.org/10.1016/j.fmre.2021.03.002>
5. Lin, W.-H., Zhong, J.-X., Liu, S., Li, T., Li, G.: Roimix: proposal-fusion among multiple images for underwater object detection. In: *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 2588–2592 (2020). IEEE. <https://doi.org/10.1109/ICASSP40776.2020.9053829>
6. Zhang, S., Wang, T., Dong, J., Yu, H.: Underwater image enhancement via extended multi-scale retinex. *Neurocomputing* **245**, 1–9 (2017). <https://doi.org/10.1016/j.neucom.2017.03.029>
7. Xu, F., Wang, H., Peng, J., Fu, X.: Scale-aware feature pyramid architecture for marine object detection. *Neural Computing and Applications* **33**, 3637–3653 (2021). <https://doi.org/10.1007/s00521-020-05217-7>
8. Girshick, R., Donahue, J., Darrell, T., Malik, J.: Rich feature hierarchies for accurate object detection and semantic segmentation. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 580–587 (2014). <https://doi.org/10.18127/j00338486-202109-11>
9. Girshick, R.: Fast r-cnn. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 1440–1448 (2015). <https://doi.org/10.1109/ICCV.2015.169>
10. Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.-Y., Berg, A.C.: Ssd: Single shot multibox detector. In: *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*, pp. 21–37 (2016). Springer. <https://doi.org/10.1007/>

11. Redmon, J., Divvala, S., Girshick, R., Farhadi, A.: You only look once: Unified, real-time object detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 779–788 (2016). <https://doi.org/10.1109/CVPR.2016.91>
12. Li, S., Li, C., Yang, Y., Zhang, Q., Wang, Y., Guo, Z.: Underwater scallop recognition algorithm using improved yolov5. *Aquacultural Engineering* **98**, 102273 (2022). <https://doi.org/10.1016/j.aquaeng.2022.102273>
13. Shen, X., Sun, X., Wang, H., Fu, X.: Multi-dimensional, multi-functional and multi-level attention in yolo for underwater object detection. *Neural computing and applications* **35**(27), 19935–19960 (2023). <https://doi.org/10.1007/s00521-023-08781-w>
14. Guo, A., Sun, K., Zhang, Z.: A lightweight yolov8 integrating fasternet for real-time underwater object detection. *Journal of Real-Time Image Processing* **21**(2), 49 (2024). <https://doi.org/10.1007/s11554-024-01431-x>
15. Cen, Q., Zhu, Q., Wang, Y., Chen, W., Liu, S.: Yolov9-yx: lightweight algorithm for underwater target detection. *The Visual Computer*, 1–13 (2024). <https://doi.org/10.1007/s00371-024-03644-4>
16. Tu, W., Yu, H., Wu, Z., Li, J., Cui, Z., Yang, Z., Zhang, X., Wang, Y.: Yolov10-udfishnet: detection of diseased takifugu rubripes juveniles in turbid underwater environments. *Aquaculture International* **33**(1), 125 (2025). <https://doi.org/10.1007/s10499-024-01798-5>
17. Khanam, R., Hussain, M.: Yolov11: An overview of the key architectural enhancements. *arXiv preprint arXiv:2410.17725* (2024). <https://doi.org/10.48550/arXiv.2410.17725>
18. Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., Rabinovich, A.: Going deeper with convolutions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1–9 (2015). <https://doi.org/10.1109/cvpr.2015.7298594>
19. Dai, J., Qi, H., Xiong, Y., Li, Y., Zhang, G., Hu, H., Wei, Y.: Deformable convolutional networks. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 764–773 (2017). <https://doi.org/10.1109/ICCV.2017.89>
20. Chollet, F.: Xception: Deep learning with depthwise separable convolutions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1251–1258 (2017). <https://doi.org/10.1109/CVPR.2017.195>
21. Chen, J., Kao, S.-h., He, H., Zhuo, W., Wen, S., Lee, C.-H., Chan, S.-H.G.: Run, don't walk: chasing higher flops for faster neural networks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 12021–12031 (2023). <https://doi.org/10.1109/CVPR52729.2023.01157>
22. Zhang, X., Liu, C., Yang, D., Song, T., Ye, Y., Li, K., Song, Y.: Rfaconv: Innovating spatial attention and standard convolutional operation. *arXiv preprint arXiv:2304.03198* (2023). <https://doi.org/10.48550/arXiv.2304.03198>
23. Zhang, X., Song, Y., Song, T., Yang, D., Ye, Y., Zhou, J., Zhang, L.: Ldconv: Linear deformable convolution for improving convolutional neural networks. *Image and Vision Computing* **149**, 105190 (2024). <https://doi.org/10.1016/j.imavis.2024.105190>
24. Hu, J., Shen, L., Sun, G.: Squeeze-and-excitation networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 7132–7141 (2018). <https://doi.org/10.1109/CVPR.2018.00745>
25. Woo, S., Park, J., Lee, J.-Y., Kweon, I.S.: Cbam: Convolutional block attention module. In: Proceedings of the European Conference on Computer Vision (ECCV), pp. 3–19 (2018). https://doi.org/10.1007/978-3-030-01234-2_1

26. Hou, Q., Zhou, D., Feng, J.: Coordinate attention for efficient mobile network design. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 13713–13722 (2021). <https://doi.org/10.1109/CVPR46437.2021.01350>
27. Wang, Q., Wu, B., Zhu, P., Li, P., Zuo, W., Hu, Q.: Eca-net: Efficient channel attention for deep convolutional neural networks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 11534–11542 (2020). <https://doi.org/10.1109/CVPR42600.2020.01155>
28. Wan, D., Lu, R., Shen, S., Xu, T., Lang, X., Ren, Z.: Mixed local channel attention for object detection. Engineering Applications of Artificial Intelligence **123**, 106442 (2023). <https://doi.org/10.1016/j.engappai.2023.106442>
29. Yu, Y., Zhang, Y., Cheng, Z., Song, Z., Tang, C.: Mca: Multidimensional collaborative attention in deep convolutional neural networks for image recognition. Engineering Applications of Artificial Intelligence **126**, 107079 (2023). <https://doi.org/10.1016/j.engappai.2023.107079>
30. Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., et al.: Pytorch: an imperative style. High-Performance Deep Learning Library **12** (2019)
31. Zhu, X., Hu, H., Lin, S., Dai, J.: Deformable convnets v2: More deformable, better results. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 9308–9316 (2019). <https://doi.org/10.1109/CVPR.2019.00953>
32. Liu, C., Li, H., Wang, S., Zhu, M., Wang, D., Fan, X., Wang, Z.: A dataset and benchmark of underwater object detection for robot picking. In: 2021 IEEE International Conference on Multimedia & Expo Workshops (ICMEW), pp. 1–6 (2021). IEEE. <https://doi.org/10.1109/ICMEW53276.2021.9455997>
33. Liu, C., Wang, Z., Wang, S., Tang, T., Tao, Y., Yang, C., Li, H., Liu, X., Fan, X.: A new dataset, poisson gan and aquanet for underwater object grabbing. IEEE Transactions on Circuits and Systems for Video Technology **32**(5), 2831–2844 (2021). <https://doi.org/10.1109/TCSVT.2021.3100059>
34. Jiang, L., Wang, Y., Jia, Q., Xu, S., Liu, Y., Fan, X., Li, H., Liu, R., Xue, X., Wang, R.: Underwater species detection using channel sharpening attention. In: Proceedings of the 29th ACM International Conference on Multimedia, pp. 4259–4267 (2021). <https://doi.org/10.1145/3474085.3475563>
35. Huang, H., Chen, Z., Zou, Y., Lu, M., Chen, C., Song, Y., Zhang, H., Yan, F.: Channel prior convolutional attention for medical image segmentation. Computers in Biology and Medicine **178**, 108784 (2024). <https://doi.org/10.1016/j.compbiomed.2024.108784>
36. Ouyang, D., He, S., Zhang, G., Luo, M., Guo, H., Zhan, J., Huang, Z.: Efficient multi-scale attention module with cross-spatial learning. In: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1–5 (2023). IEEE. <https://doi.org/10.1109/ICASSP49357.2023.10096516>
37. Lau, K.W., Po, L.-M., Rehman, Y.A.U.: Large separable kernel attention: Rethinking the large kernel attention design in cnn. Expert Systems with Applications **236**, 121352 (2024). <https://doi.org/10.1016/j.eswa.2023.121352>
38. Zhang, Q.-L., Yang, Y.-B.: Sa-net: Shuffle attention for deep convolutional neural networks. In: ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 2235–2239 (2021). IEEE. <https://doi.org/10.1109/ICASSP39728.2021.9414568>
39. Wang, B., Wang, Z., Guo, W., Wang, Y.: A dual-branch joint learning network for underwater object detection. Knowledge-Based Systems **293**, 111672 (2024). <https://doi.org/10.1016/j.knosys.2024.111672>

40. Ren, S., He, K., Girshick, R., Sun, J.: Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in neural information processing systems* **28** (2015). <https://doi.org/10.1109/TPAMI.2016.2577031>
41. Cai, Z., Vasconcelos, N.: Cascade r-cnn: Delving into high quality object detection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 6154–6162 (2018). <https://doi.org/10.1109/CVPR.2018.00644>
42. Zhao, Y., Lv, W., Xu, S., Wei, J., Wang, G., Dang, Q., Liu, Y., Chen, J.: Detsr beat yolos on real-time object detection. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16965–16974 (2024). <https://doi.org/10.48550/arXiv.2304.08069>
43. Peng, Y., Li, H., Wu, P., Zhang, Y., Sun, X., Wu, F.: D-fine: Redefine regression task of detsr as fine-grained distribution refinement. In: Yue, Y., Garg, A., Peng, N., Sha, F., Yu, R. (eds.) *International Conference on Representation Learning*, vol. 2025, pp. 44015–44031 (2025). <https://doi.org/10.48550/arXiv.2410.13842>
44. Song, P., Li, P., Dai, L., Wang, T., Chen, Z.: Boosting r-cnn: Reweighting r-cnn samples by rpn’s error for underwater object detection. *Neurocomputing* **530**, 150–164 (2023). <https://doi.org/10.1109/ICCV48922.2021.00349>
45. Liang, X., Song, P.: Excavating roi attention for underwater object detection. In: *2022 IEEE International Conference on Image Processing (ICIP)*, pp. 2651–2655 (2022). IEEE. <https://doi.org/10.48550/arXiv.2206.12128>
46. Dai, L., Liu, H., Song, P., Liu, M.: A gated cross-domain collaborative network for underwater object detection. *Pattern Recognition* **149**, 110222 (2024). <https://doi.org/10.1016/j.patcog.2023.110222>
47. Fu, C., Fan, X., Xiao, J., Yuan, W., Liu, R., Luo, Z.: Learning heavily-degraded prior for underwater object detection. *IEEE Transactions on Circuits and Systems for Video Technology* **33**(11), 6887–6896 (2023) <https://doi.org/10.1109/TCSVT.2023.3271644>
48. Dai, L., Liu, H., Song, P., Tang, H., Ding, R., Li, S.: Edge-guided representation learning for underwater object detection. *CAAI Transactions on Intelligence Technology* **9**(5), 1078–1091 (2024) <https://doi.org/10.1049/cit2.12325>
49. Jocher, G.: YOLOv5 by Ultralytics. <https://doi.org/10.5281/zenodo.3908559> . <https://github.com/ultralytics/yolov5>
50. Li, C., Li, L., Jiang, H., Weng, K., Geng, Y., Li, L., Ke, Z., Li, Q., Cheng, M., Nie, W., Li, Y., Zhang, B., Liang, Y., Zhou, L., Xu, X., Chu, X., Wei, X., Wei, X.: Yolov6: A single-stage object detection framework for industrial applications. *ArXiv abs/2209.02976* (2022). <https://api.semanticscholar.org/CorpusID:252110986>
51. Wang, C.-Y., Bochkovskiy, A., Liao, H.-Y.M.: Yolov7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 7464–7475 (2023). <https://doi.org/10.48550/arXiv.2207.02696>
52. Jocher, G., Chaurasia, A., Qiu, J.: Ultralytics YOLOv8. <https://github.com/ultralytics/ultralytics>
53. Wang, C.-Y., Yeh, I.-H., Mark Liao, H.-Y.: Yolov9: Learning what you want to learn using programmable gradient information. In: *European Conference on Computer Vision*, pp. 1–21 (2024). Springer. https://doi.org/10.1007/978-3-031-72751-1_1
54. Wang, A., Chen, H., Liu, L., Chen, K., Lin, Z., Han, J., *et al.*: Yolov10: Real-time end-to-end object detection. *Advances in Neural Information Processing Systems* **37**, 107984–108011 (2025). <https://doi.org/10.48550/arXiv.2405.14458>
55. Tian, Y., Ye, Q., Doermann, D.: Yolov12:

Attention-centric real-time object detectors.
arXiv preprint arXiv:2502.12524 (2025).
<https://doi.org/10.48550/arXiv.2502.12524>

56. Lei, M., Li, S., Wu, Y., Hu, H., Zhou, Y., Zheng, X., Ding, G., Du, S., Wu, Z., Gao, Y.: YOLOv13: Real-Time Object Detection with Hypergraph-Enhanced Adaptive Visual Perception (2025). <https://arxiv.org/abs/2506.17733>
57. Ouyang, J., Li, Y.: Enhanced underwater object detection via attention mechanism and dilated large-kernel networks. *The Visual Computer*, 1–23 (2025). <https://doi.org/10.1007/s00371-025-03870-4>

ARTICLE IN PRESS